PAPER ID-310477

Subject Code: KCE078

Roll No: | | | | | | | | | | | | | |

BTECH

(SEM VII) THEORY EXAMINATION 2023-24

IRRIGATION AND WATER RESOURCE ENGINEERING

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question
a.	Describe Probable Maximum Precipitation (PMP).
b.	What is the assumption made in unit hydrograph?
c.	Discuss Horton's infiltration equation.
d.	What is sediment rating curve?
e.	Define trickle irrigation system.
f.	What is an oulet in irrigation?
g.	Define the objectives of diversion headwork.
h.	Write the methods of bank protection.
i.	Define infiltration gallery.
j.	A well is to constructed in a fine sandy subsoil formation. The discharge of the well is anticipated to be 0.004 m3/s under the depression head of 4 m. find out the diameter of the well.

SECTION B

2. Attempt any *three* of the following: **10x3=30**

a.	Describe the factors that affect the process of infiltration. How will you measure the rate of infiltration?
b.	Give the steps of derivation of unit hydrograph.
c.	Design a concrete lined channel of triangular section to carry a discharge of 45 m^3/sec at a slope of 1 in 1000. The side slopes of the channel are 1.5:1 and Manning's rugosity coefficient for lining material as 0.018.
d.	Explain various types of canals.
e.	A tube well penetrates into an unconfined aquifer having a saturated depth of 80 m. When the drawdown was 12 m discharge from the well was 180 liters per minute, Assume steady state condition and radius of circle of influence same for both cases. Calculate the discharge at 18 m drawdown.

SECTION C

3. Attempt any *one* part of the following: **10x1=10**

a.	How to estimate evaporation from free water surface by Meyer's formula.
b.	Draw a sketch of recording type rain-gauge and explain its working.

4. Attempt any *one* part of the following: **10x1=10**

a.	Describe the various method of irrigation system. Define sprinkler irrigation system with neat sketch.

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b.	Water course has a culturable commanded area of 1200 hectares. The intensity of irrigation for crop A is 40 % and for B is 35 % both the crops being Rabi crops. Crop A has a kor period of 20 days and crop B has kor period of 15 days. Calculate the discharge of the water course if the kor depth for crop A is 10 cm and for B is 16 cm.

5. Attempt any *one* part of the following: 10x1=10

a.	Using Lacey's theory, design a trapezoidal irrigation channel (side slope, 1H:2V) carrying discharge of 40 m^3/sec. Take silt factor as 1.0.
b.	Explain with typical cross-section of a surface drain.

6. Attempt any *one* part of the following: 10x1=10

a.	Describe the functions of various component parts of a Diversion head work.
b.	Explain requirements of irrigation outlet. Draw a neat sketch of Crump's proportional module and explain its working.

7. Attempt any *one* part of the following: 10x1=10

a.	Give the methods of development of tube wells. Explain with neat sketch of development by high velocity jetting.
b.	What do you meant by interference of wells. Explain with neat sketch.

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